

Lecture #8

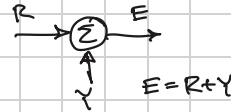
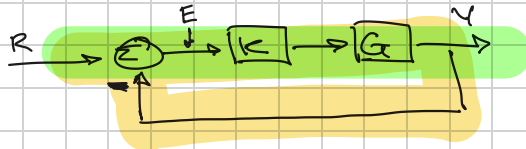
Homework #1 due today

Homework #2 due 8 Oct. '26

Office Hours T/Th 2-3:30 EMA205

Block Diagrams

$$E = R - Y$$



$$T(s) = \frac{Y(s)}{R(s)} = \frac{\text{FORWARD}}{1 + \text{LOOP}} = \frac{KG}{1 + KG}$$

Warm Up:

$$G(s) = \frac{1}{s + \alpha} \quad (\text{"Plant"})$$

$$K(s) = K_P + \frac{K_I}{s} \quad (\text{"controller"})$$

(P) (I)

$$= \frac{K_P s + K_I}{s} \rightarrow \begin{array}{l} \text{one zero} \\ \text{one pole (at the origin)} \end{array}$$

Calculate $T(s) = \frac{Y(s)}{R(s)}$

① Calculate $K(s) \cdot G(s)$

$$= \frac{sK_P + K_I}{s(s + \alpha)}$$

② $T(s) = \frac{KG}{1 + KG} = \frac{\left(\frac{sK_P + K_I}{s(s + \alpha)} \right)}{1 + \left(\frac{sK_P + K_I}{s(s + \alpha)} \right)}$

$$= \frac{sK_P + K_I}{s(s + \alpha) + (sK_P + K_I)} = T(s)$$

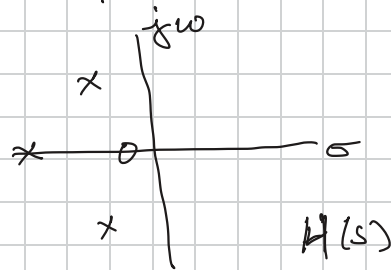
Q: For the plant $G(s)$ w/ feedback control by a PI controller, what is the closed-loop transfer fun?

① what is the numerator? *zeros of KG*

② what about the denominator? *poles are different under feedback*

Today's Lecture: Time-domain Spec's

"Looking at the Matrix"



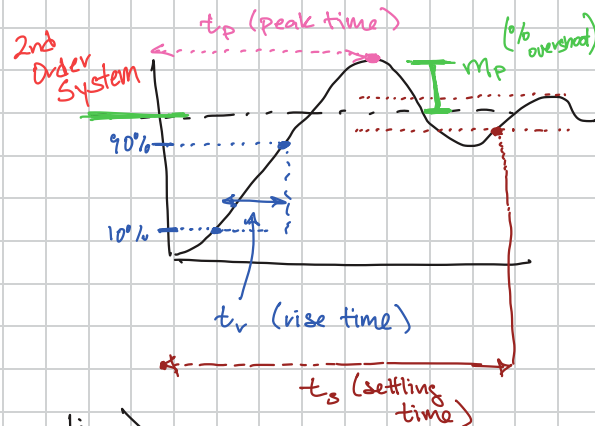
- How to infer time-domain characteristics from pole and zero locations.

- Today
- 1) What are the specs?
 - 2) How do we do design synthesis from the spec's?
 - 3) What zeroes do to the response?
 - 4) What multiple poles do to the response?
 - 5) Demo

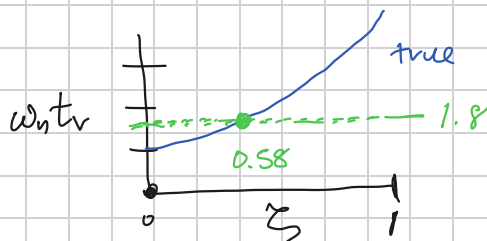
The four spec's :

- 1) Rise time
- 2) Overshoot (peak time)
- 3) Settling

Context : 2nd Order System



① Rise time : $t_r \approx \frac{1.8}{\omega_n}$ (approximation)



② Overshoot : $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 u$
(and peak time)

Sol'n: $y(t) = 1 - e^{-\sigma t} \left(\cos(\omega_d t) + \frac{\sigma}{\omega_d} \sin(\omega_d t) \right)$

$$\omega_d \triangleq \omega_n \sqrt{1-\zeta^2}$$

$$\sigma \triangleq \omega_n \zeta$$

$y(t) = 0$ solve for t to find t_p

when we have t_p , then we evaluate $\frac{y(t_p) - y(\infty)}{y(\infty)} = m_p$

$$\dot{y} = \frac{\omega_n^2}{\omega_d} \underbrace{e^{-\sigma t}}_{>0} \underbrace{\sin(\omega_d t)}_{=0} = 0$$

$$\Rightarrow \sin(\omega_d t) \equiv 0$$

$$\Rightarrow t_p = \frac{\pi}{\omega_d} \quad (\text{exact})$$

what m_p ? $y(t_p) = 1 + e^{-\sigma\pi/\omega_d}$

$y(\infty) = 1$ (FVT)

$$m_p = \frac{y(t_p) - y(\infty)}{y(\infty)} = e^{-\sigma\pi/\omega_d} \quad (\text{exact})$$

$$\Rightarrow m_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$\sigma = \omega_n \zeta$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

m_p only depends ζ ; doesn't have ANY dependence on ω_n .

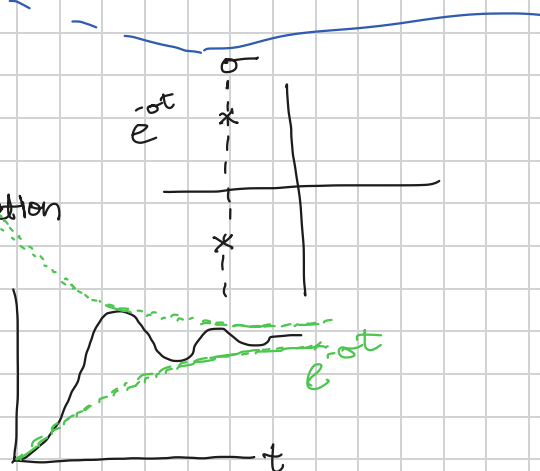
3) Settling Time

1% settling time: $t_s \approx \frac{4.6}{\zeta\omega_n}$ (approx.)

$$\sigma = \zeta\omega_n$$

↑ dominant real pole location

$$\sigma \pm j\omega$$



What about 1st Order Systems?

$$H(s) = \frac{\sigma}{s + \sigma} \Rightarrow y(t) = (1 - e^{-\sigma t}) 1(t) \quad \begin{array}{l} \text{driven by} \\ \text{real part} \\ \text{of pole} \end{array}$$

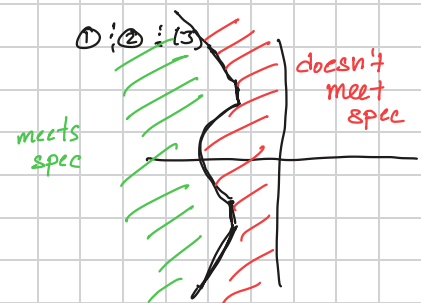
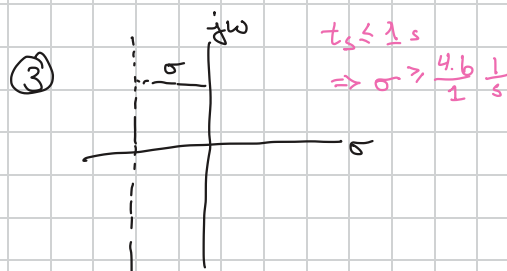
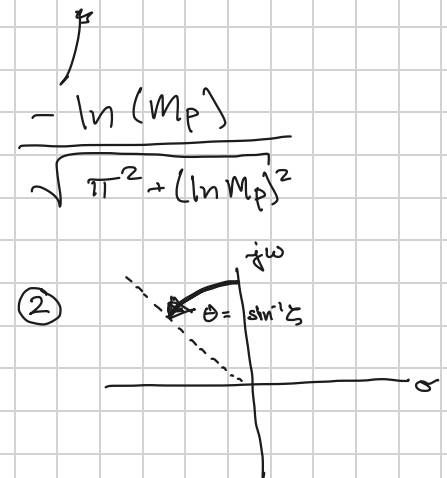
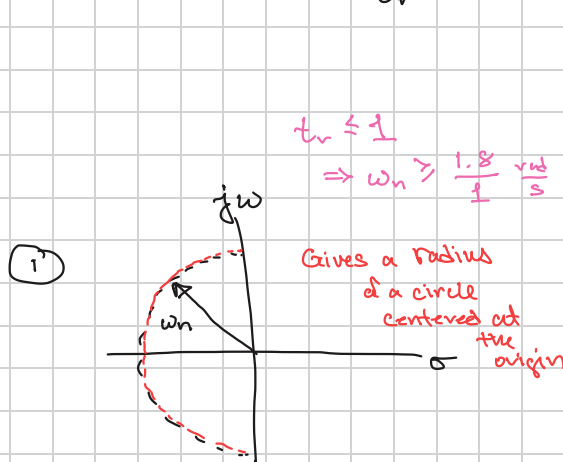
(step response)

$$\boxed{M_p = \emptyset} \quad \boxed{t_r = \frac{\ln 9}{\sigma}} \quad \boxed{t_s = \frac{\ln 100}{\sigma}} \quad \boxed{\tau = \frac{1}{\sigma}}$$

↑
there is no t_p b/c
the system doesn't
have a peak.

Design Synthesis : Given quantities like overshoot, rise time, etc. Where can/should the poles (and zeros) be located.

$$\textcircled{1} \quad \omega_n \geq \frac{1.8}{t_r} \quad \textcircled{2} \quad \zeta \geq \zeta(M_p) \quad \textcircled{3} \quad \sigma \geq \frac{4.6}{t_s}$$



What happens when we add a zero to the TF?

$$H_1(s) = \frac{2}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{2}{s+2} = 2e^{-t} - 2e^{-2t}$$

$$H_2(s) = \frac{2(s+1.1)}{1.1(s+1)(s+2)} = \frac{2/11}{s+1} + \frac{18/11}{s+2} = \frac{2}{11}e^{-t} + \frac{18}{11}e^{-2t}$$

The modes are the same, but
the zero weighted the modes
differently.

Normalized Family:

$$H(s) = \frac{\overset{\alpha\text{-term}}{\downarrow} (s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \quad \begin{matrix} \swarrow 1 \text{ zero} \\ \nwarrow 2 \text{ poles} \end{matrix}$$

$\alpha > 0$ is the ratio of the zero's distance
from the imag. axis to the pair's dist. ω

Large $\alpha \Rightarrow$ the LHP zero is far away from poles

Small $\alpha \Rightarrow$ the LHP zero is close to the poles

Zeros "look" like derivatives (and a zero at the origin is a derivative)

$$H(s) = \frac{\frac{s}{\alpha\zeta} + 1}{s^2 + 2\zeta\omega_n s + 1}$$

$$= \underbrace{\frac{1}{s^2 + 2\zeta\omega_n s + 1}}_{H_0(s) \text{ (no zeros)}} + \underbrace{\frac{1}{\alpha\zeta}}_{\text{scale}} \underbrace{\frac{s}{s^2 + 2\zeta\omega_n s + 1}}_{s \cdot H_0(s)}$$

If our output is $y(t)$, then we know it

LHP

$$y(t) = y_0(t) + \frac{1}{\alpha\zeta} \dot{y}_0(t)$$

↓
response of $H_0(s)$ (no zeros)

added the derivative term

What happens when zeros are in RHP?

→ "Nonminimum Phase Zeros"

RHP

$$y(t) = y_0(t) - \frac{1}{|\alpha|\zeta} \dot{y}_0(t)$$

"undershoot"

MP zero

NMP zero

causes undershoot